Agnostic Learnability of Halfspaces via Logistic Loss

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Setting.

Binary classification: given by distribution P on $\mathbb{R}^d \times \{-1,+1\}$; we have i.i.d. samples from P.
Linear classifier: $x \mapsto \langle x, w\rangle$.
Goal: compete with the optimal solution $\bar{u}$, where

$$\mathcal{R}_{0-1}(\bar{u}) := \Pr_{(x,y)\sim P}\left(\mathrm{sign}(\langle \bar{u}, x\rangle) \neq y\right) = \mathrm{OPT} > 0.$$

Logistic regression (LR): let $\ell_{\log}(z) := \ln(1+e^{-z})$, and let

$$\mathcal{R}_{\log}(w) := \mathbb{E}_{(x,y)\sim P}\left[\ell_{\log}\left(y\langle w, x\rangle\right)\right]$$

denote the population logistic risk of w over P.

Contributions.

Prior lower and upper bounds for LR do not match:

- General P: $1-\mathrm{OPT}$ lower bound (Ben-David et al., 2012).
- Isotropic log-concave distributions: $\widetilde{\Omega}(\mathrm{OPT})$ lower bound (Diakonikolas et al., 2020).
- "Well-behaved" sub-exponential distributions: $\widetilde{O}(\sqrt{\mathrm{OPT}})$ upper bound with SGD (Frei et al., 2021).

On "well-behaved" conditions:

- Standard concentration and anti-concentration conditions.
- Example: a mixture of log-concave dist. (e.g., Gaussian).

Our lower and upper bounds for LR:

- $\Omega(\sqrt{\mathrm{OPT}})$ lower bound for well-behaved sub-exp dist.; matching $\widetilde{O}(\sqrt{\mathrm{OPT}})$ upper bound from (Frei et al., 2021).
- $\widetilde{O}(\mathrm{OPT})$ upper bound with "radial Lipschitzness."

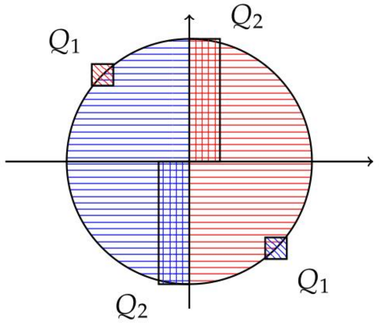
Upper bounds beyond LR:

- Diakonikolas et al. (2020) designed a nonconvex SGD that achieves $O(\mathrm{OPT})+\epsilon$ risk using $\widetilde{O}(d/\epsilon^4)$ samples. They can also handle heavy-tailed distributions.
- Other prior algorithms achieving $O(\mathrm{OPT})+\epsilon$ risk involve solving multiple rounds of convex program (Awasthi et al., 2014; Daniely, 2015).
- We design a simple two-phase convex program that achieves $O(\mathrm{OPT}\ln(1/\mathrm{OPT}))+\epsilon$ risk using $\widetilde{O}(d/\epsilon^2)$ samples, **without** radial Lipschitzness.

Lower bound.

We construct a distribution Q with three parts Q_1, Q_2, Q_4. (One part Q_3 is omitted for simplicity.)

1. Q_1 consists of two squares with edge length $\sqrt{\mathrm{OPT}/2}$ and density 1; one has center $(\sqrt{2}/2, -\sqrt{2}/2)$ and label -1, the other has center $(-\sqrt{2}/2, \sqrt{2}/2)$ and label $+1$.
2. Q_2 is supported on $[0,\sqrt{\mathrm{OPT}}]\times[0,1]$ and $[-\sqrt{\mathrm{OPT}},0]\times[-1,0]$ with density 1, and the label is given by $\mathrm{sign}(x_1)$.
3. Q_4 is the uniform distribution over the unit ball with density $q_4 := \frac{1-\mathrm{OPT}-2\sqrt{\mathrm{OPT}}}{\pi}$ and label $\mathrm{sign}(x_1)$.



Theorem. Q is well-behaved and sub-exponential, and the global minimizer w^* of $\mathcal{R}_{\log}$ has $\mathcal{R}_{0-1}(w^*) = \Omega(\sqrt{\mathrm{OPT}})$.

Proof ideas:

- To minimize $\mathcal{R}_{\log}$ on Q_4, we should go to infinity along $(1,0)$.
- Q_1 keeps w^* bounded with $\|w^*\| = O(1/\sqrt{\mathrm{OPT}})$.
- Q_2 drags w^* above by an angle of $\Omega(\sqrt{\mathrm{OPT}})$.

Upper bound: radial Lipschitzness.

To exclude Q and overcome the lower bound, we make the following assumption for LR.

Radial Lipschitzness: There exists a measurable function $\kappa : \mathbb{R}_+ \to \mathbb{R}_+$ such that for any two-dimensional subspace V, letting p_V denote the density of the projection of feature distribution onto V, then

$$|p_V(r,\theta) - p_V(r,\theta')| \le \kappa(r)|\theta-\theta'|.$$

- Holds if p_V is Lipschitz, even non-log-concave ones (e.g., Gaussian mixtures).
- Does not hold for general log-concave distributions.
- Does not hold for Q, non-Lipschitz near the vertical axis.

Upper bound: logistic regression.

Projected gradient descent (PGD): let $w_0 := 0$, and

$$w_{t+1} := \Pi_{\mathcal{B}(1/\sqrt{\epsilon})}\left[w_t - \eta\nabla\widehat{\mathcal{R}}_{\log}(w_t)\right].$$

Theorem. If the dist. is well-behaved, sub-exponential and radially-Lipschitz, with learning rate $\widetilde{\Theta}(1/d)$, using $\mathrm{poly}(d, 1/\epsilon, \ln(1/\delta))$ samples and iterations, with prob. $1-\delta$, PGD outputs w_t with

$$\mathcal{R}_{0-1}(w_t) = \widetilde{O}(\mathrm{OPT}) + \epsilon.$$

Proof ideas: under the given conditions, suppose $\hat{w}$ satisfies $\mathcal{R}_{\log}(\hat{w}) \le \mathcal{R}_{\log}(\|\hat{w}\|\bar{u}) + \epsilon'$, then

$$\mathcal{R}_{0-1}(\hat{w}) = \widetilde{O}\left(\max\left\{\mathrm{OPT}, \sqrt{\frac{\epsilon'}{\|\hat{w}\|}}, \frac{C_\kappa}{\|\hat{w}\|^2}\right\}\right). \quad (1)$$

- $C_\kappa = O(\ln(1/\mathrm{OPT})^2)$ for Lipschitz density.
- The global optimizer w^* of $\mathcal{R}_{\log}$ satisfies $\|w^*\| = \widetilde{\Omega}(1/\sqrt{\mathrm{OPT}})$; PGD also ensures w_t has small ϵ' and $\|w_t\| = \widetilde{\Omega}(1/\sqrt{\mathrm{OPT}})$.

Upper bound: two-phase convex program.

Motivation: (1) holds for the hinge loss $\ell_h(z) := \max\{-z, 0\}$ **without** radial Lipschitzness!

$$\mathcal{R}_{0-1}(\hat{w}) = \widetilde{O}\left(\max\left\{\mathrm{OPT}, \sqrt{\frac{\epsilon'}{\|\hat{w}\|}}\right\}\right).$$

However, the global minimizer of $\mathcal{R}_h$ is 0; if we find $\hat{w}$ by global minimization, as ϵ' gets small, $\|\hat{w}\|$ also gets small.

Ideas: LR followed by restricted Perceptron:

- LR finds unit v that is $\widetilde{O}(\sqrt{\mathrm{OPT}})$ away from $\bar{u}$.
- Then minimize $\mathcal{R}_h$ over $\mathcal{D} := \{w | \langle w, v\rangle \ge 1\}$.
  - $\forall w \in \mathcal{D}$, $\|w\| \ge 1$.
  - $\|\hat{w}\|\bar{u}$ may not in $\mathcal{D}$, but $\left(1+\widetilde{O}(\mathrm{OPT})\right)\|\hat{w}\|\bar{u} \in \mathcal{D}$!
- Use SGD for better sample complexity.

Theorem. If the dist. is well-behaved and sub-exponential, using $\widetilde{O}(d/\epsilon^2)$ samples, SGD can achieve zero-one risk $O(\mathrm{OPT}\ln(1/\mathrm{OPT}))+\epsilon$.